A seven-T fixed measurement accessing 220 four-qubit Pauli directions

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The non-Clifford gate cost of a single fixed quantum-state measurement is a finite-resource question distinct from the number of available outcomes. We give an explicit Clifford+T circuit on four input qubits and seven pure stabilizer ancillas whose complete, 2048-outcome measurement has an exact operator-span rank of 220. The circuit uses seven T or $T^\dagger$ gates. Its rank follows from an exact Pauli-branch enumeration: 260 nonzero contracted readout operators occupy 220 distinct four-qubit Pauli directions. This exceeds the rank 208 attained by a particular previously reported seven-T construction at the same input size and gate count. The remaining 36 blind directions make our measurement informationally incomplete. The example gives a stronger attainable partial-information benchmark; it does not determine the maximum seven-T rank or the minimum gate count for complete four-qubit tomography.

I. INTRODUCTION

A fixed generalized measurement can extract information from an unknown quantum state without changing the circuit between shots. Single-setting tomography has been demonstrated using enlarged local Hilbert spaces and symmetric informationally complete readout [1]. For qubit circuits restricted to stabilizer ancillas and Clifford+T gates, the non-Clifford resource cost is a different question. Lo Monaco *et al.* [2] give general bounds and constructions for the rank of the measurement frame. Their Theorem 7 supplies an *attainable* rank of 208 for four input qubits and seven T gates; it is not an upper bound on what another circuit might achieve.

Here we exhibit one physical circuit at that same input size and T count with rank 220. The improvement is in the dimension of the exactly accessible operator space, not in finite-shot accuracy, noise tolerance, Clifford depth, or complete tomography. In particular, the unresolved question of whether any circuit with at most seven T gates can reach the full four-qubit rank 256 remains untouched by this construction.

II. MEASUREMENT MODEL AND EXPLICIT CIRCUIT

Let D consist of four arbitrary input qubits and A of seven ancillas prepared in $|0^7\rangle$. All 11 qubits are measured at the end in the computational basis; there is no intermediate readout, postselection, or randomized circuit choice. For a unitary U and $b \in \{0,1\}^{11}$, the effective data effect is

$$E_b = (I_D \otimes \langle 0^7|)U^\dagger |b\rangle\langle b| U(I_D \otimes |0^7\rangle). \tag{1}$$

Thus $E_b \succeq 0$ and $\sum_b E_b = I_D$. On the 256-dimensional real space of Hermitian data operators, the measurement frame satisfies

$$\begin{aligned} F &= \sum_b |E_b\rangle\!\rangle\langle\!\langle E_b|, \\ \operatorname{rank} F &= \dim_{\mathbb{R}} \operatorname{span}\{E_b\}. \end{aligned} \tag{2}$$

Number the data wires $0,\dots,3$ and ancilla wires $4,\dots,10$. Binary masks $x, z \in \{0,\dots,2047\}$ use bit i for wire i, least significant bit first. Define the Hermitian Pauli

$$\begin{aligned} W(x,z) &= i^{\mathrm{wt}(x\&z)} \prod_{i=0}^{10} X_i^{x_i} Z_i^{z_i}, \\ \omega((x,z),(x',z')) &= x\cdot z' + z\cdot x' \pmod 2. \end{aligned} \tag{3}$$

The seven rotations $R_j = \exp[-is_j \pi W(x_j, z_j)/8]$ act in increasing j order and have the exact masks in Table I. Each is a Clifford conjugate of T or $T^\dagger$, up to an irrelevant global phase; $T^\dagger = S^\dagger T$ costs one non-Clifford gate in this convention.

Following the rotations, a single stabilizer readout measures the maximal commuting group generated by the eleven Pauli labels in Table II. The labels are mutually commuting and binary independent. A Clifford C maps the computational Z_i generators to these labels, up to signs, so the actual terminal circuit is

$$U = C^\dagger R_7 R_6 \cdots R_1. \tag{4}$$

The accompanying witness specifies a chronological 132-operation $H/S/\mathrm{CX}$ realization of C and therefore removes any synthesis ambiguity. An explicit Clifford+T expansion has 333 gates, exactly seven of them T or $T^\dagger$. The Clifford count is not claimed minimal.

III. EXACT RANK CERTIFICATE

The group generated by Table II contains 2^{11} Hermitian Paulis Q with eigenvalue-sign conventions fixed by C. The Fourier transformation between the projectors of this complete commuting readout and its group elements

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TABLE I. Seven chronological Pauli rotations. The sign s_j selects T or $T^\dagger$ up to Clifford factors and global phase. Masks use the convention of Eq. (3).

j	x_j	z_j	s_j	j	x_j	z_j	s_j
1	837	805	$+1$	5	636	1340	-1
2	1194	801	$+1$	6	1562	1695	$+1$
3	120	1509	$+1$	7	1029	124	-1
4	38	1453	-1				

TABLE II. Binary (x, z) labels of the eleven independent, commuting readout generators. Generator signs only relabel outcomes and do not affect the operator span.

i	x	z	i	x	z	i	x	z
0	1968	1249	4	985	1777	8	1851	2019
1	1901	652	5	1880	532	9	346	1556
2	539	1310	6	1880	880	10	1330	641
3	24	292	7	1285	1213			

is invertible. Consequently the span of the physical effects in Eq. (1) has the same dimension as the span of the contracted Heisenberg operators

$$A_Q = \langle 0^7 | R_1^\dagger \cdots R_7^\dagger Q R_7 \cdots R_1 | 0^7 \rangle. \tag{5}$$

For a Hermitian Pauli P and Pauli Q, conjugation by a rotation in Eq. (4) leaves Q unchanged when P and Q commute. When they anticommute, it gives

$$e^{is\pi P/8} Q e^{-is\pi P/8} = \frac{Q + isPQ}{\sqrt{2}}. \tag{6}$$

The two terms have distinct binary labels and nonzero coefficients. This elementary branch rule is enough to compute the rank exactly; no floating-point rank threshold is used.

Write a_j for the seven-bit ancillary-X mask of rotation j. The masks are

$$\begin{aligned} (a_1, \ldots, a_7) &= (52, 74, 7, 2, 39, 97, 64), \\ \operatorname{rank}_{\mathbb{F}_2} [a_1 \ \cdots \ a_7] &= 7. \end{aligned} \tag{7}$$

For a given Q, the projector $\langle 0^7| \cdot |0^7\rangle$ annihilates every Pauli branch with a nonzero ancillary-X mask. A branch selecting the product term at rotation indices $S \subseteq \{1, \ldots, 7\}$ has final ancillary-X mask $x_A(Q) + \sum_{j \in S} a_j$. Equation (7) implies that *at most one* branch subset can survive: its selection vector is the unique solution of $[a_1 \ \cdots \ a_7] e = x_A(Q)$ over $\mathbb{F}_2$. If a requested product branch is unavailable because its current Pauli commutes with the corresponding axis, then $A_Q = 0$. Otherwise A_Q is a nonzero real multiple of a single four-qubit Pauli. Hence cancellation between surviving branches is impossible.

We enumerate all 2^{11} readout-group masks and follow each unique branch in the reverse rotation order $j = 7, \ldots, 1$. The Pauli update on a selected product branch is $(x, z) \leftarrow (x, z) + (x_j, z_j)$ over $\mathbb{F}_2$; it is allowed exactly when the current symplectic pairing with (x_j, z_j) is one. Unselected branches leave the label unchanged and have a nonzero coefficient whether they commute or anticommute. This finite exact calculation produces the counts in Table III. The accompanying code and JSON record every resulting label, including all 36 absent data labels. Because distinct four-qubit Paulis are real-linearly independent, the 220 distinct nonzero labels prove the claimed rank.

TABLE III. Exact enumeration for the circuit of Tables I and II. The 260 surviving Fourier rows need not yield distinct data operators.

Readout-group elements and physical outcomes	2048
Vanishing contracted Fourier rows	1788
Nonzero contracted Fourier rows	260
Accessible data-Pauli labels (rank)	**220**
Absent data-Pauli labels	36

An independent state-vector construction applies the synthesized gate sequence to all 16 embedded data basis states in the exact field $\mathbb{F}_{1009}$, where the required images of i and $\sqrt{2}$ exist. It checks the isometry and $\sum_b E_b = I_{16}$, obtains effect-matrix rank 220, and gives a nonzero 220-by-220 minor (determinant 955 modulo 1009). This supplies an independent lower-rank check; the unique-branch argument supplies the characteristic-zero upper bound. The modular calculation alone would not certify the upper bound.

OpenAI Codex (model configuration `gpt-5.6-sol`, ultra reasoning setting) substantively assisted the circuit search, exact-certificate code generation, and the computational checks. Separate AI-assisted implementations reconstructed the primary two-T source example, enu-

merated all 2048 contracted operators, and evolved the explicit state matrix over the finite field. Their inputs, algorithms, and outputs are supplied for human scrutiny; the independent implementations are not a substitute for human scientific review.

IV. COMPARISON AND LIMITS

For the same $n = 4$ and $t = 7$, the constructive value of Theorem 7 in Ref. [2] is

$$B(4, 7) = 2^{-3}(3^3 - 1)(3^2 - 1)^2 = 208. \tag{8}$$

That theorem presents 208 as achievable, while leaving the maximum open. The present rank 220 raises the demonstrated attainable rank by 12 independent data-Pauli directions relative to that particular construction. Rank is invariant under unitary Clifford relabeling of the input and invertible relabeling of outcomes, so the 220-dimensional span cannot be equivalent in rank to its 208-dimensional span. This comparison makes no claim of global priority beyond the cited work or of optimality among seven-T circuits.

The 36 missing directions imply that some distinct four-qubit density operators give the same ideal outcome distribution. The exact rank says nothing about the smallest nonzero frame eigenvalue, statistical variance, gate noise, finite-shot reconstruction, or hardware connectivity. The construction uses seven ancillas and a long Clifford network; neither is optimized. A rank-256 measurement might still exist with seven T gates and a different circuit or ancilla count, and this example does not resolve that possibility.

DATA AND SOFTWARE AVAILABILITY

The exact witness, the branch-enumeration program and result record, and the independent explicit-circuit replay and result record accompany this draft in its reproducibility source packet. The data are finite integers and exact-field outputs; no experimental data were used. A persistent public archive and its citation are pending an author decision before submission.

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OpenAI Codex (model configuration `gpt-5.6-sol`, ultra reasoning setting) also assisted source comparison and manuscript drafting. The substantive research use and computational checks are described in Sec. III. These records are available for human author assessment; no claim of completed human or external peer review is made.

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